AI Copilot for Manufacturing Process Planning and Quality Inspection

25ARE471 Introduction to Generative AI
End-Semester Project Presentation

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Motivation / Pain Points

Fragmented Expertise

Manufacturing process planning depends heavily on experienced engineers and scattered shop-floor knowledge.

Siloed Inspection Data

Quality inspection produces images, measurements and reports that are often analyzed in separate systems.

Time-Consuming Planning

Creating routing sheets, operation sequences, tool selections and inspection plans is time-consuming.

Goal

Combine Generative AI + vision + retrieval to support faster, traceable engineering decisions.

Problem Definition / Objectives

Problem

Build a GenAI copilot that converts part requirements and plant knowledge into a draft process plan and assists with visual quality inspection.

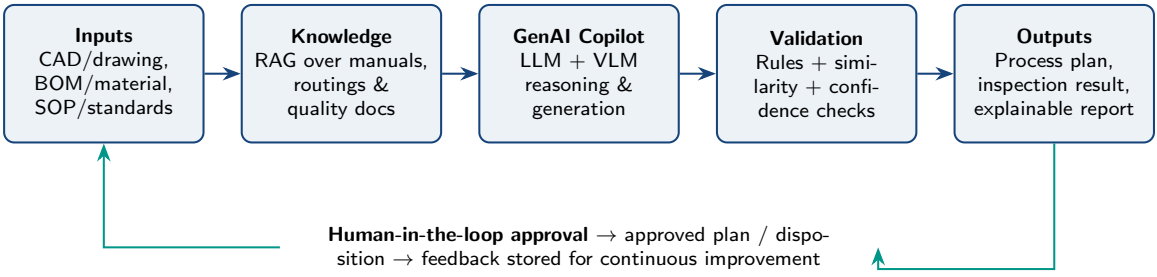
Objective 1. Generate operation sequence, machine/tool suggestions and inspection checkpoints using retrieved engineering knowledge.

Objective 2. Analyze inspection images with a vision model and explain likely defects with confidence and evidence.

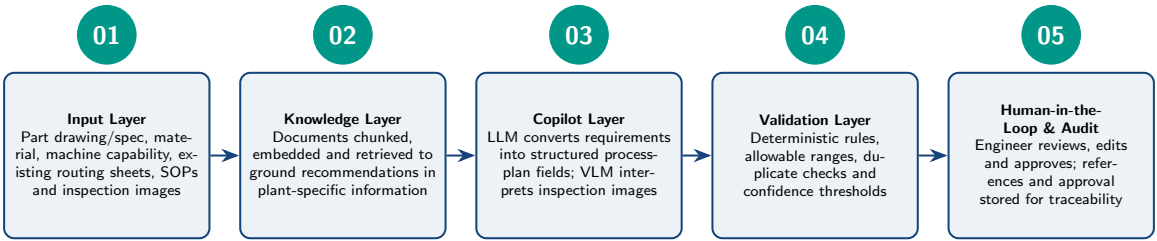
Objective 3. Provide traceable recommendations, human approval and structured outputs instead of unrestricted text generation.

Success Criteria. Useful draft plans · defect detection quality · grounded answers · reduced planning effort · human acceptance.

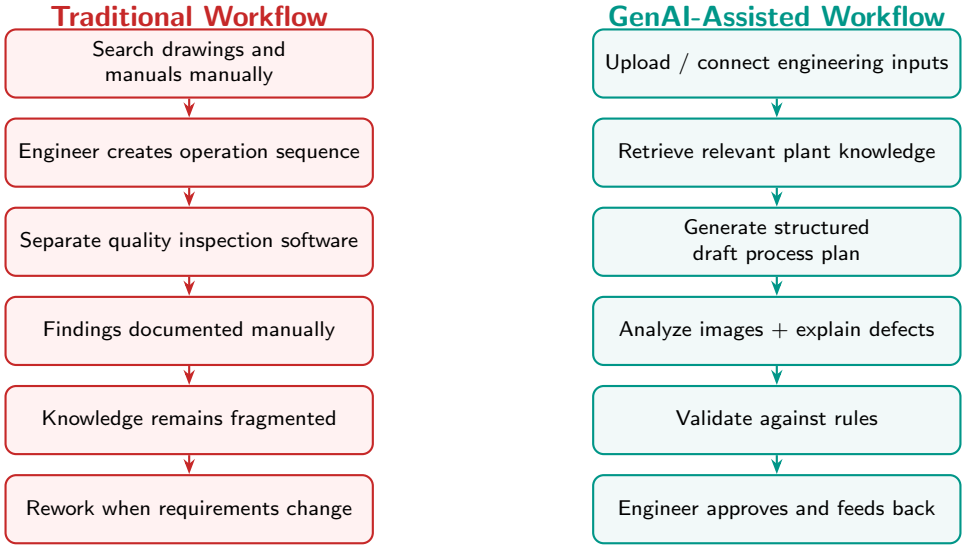
Solution Formulation



Solution Formulation – Explanation



Traditional Workflow vs GenAI Workflow



Usage of GenAI Tools

LLM

- Generate process-plan drafts
- Explain recommendations
- Convert free text $\rightarrow$ JSON

Vision Model

- Inspect part images
- Classify / localize defects
- Describe visual evidence

RAG & Vector DB

- Retrieve SOPs, standards, routings and lessons learned
- Store embeddings & similarity search
- Source traceability

Python Stack

- Orchestration · APIs · Evaluation
- OpenCV · LangChain/LlamaIndex-style RAG
- FAISS/Chroma vector store · FastAPI/Streamlit interface

Challenge Identification

Challenge	Mitigation
1. Hallucination / unsupported advice	RAG, source citations, constrained output schema, human approval
2. Poor image quality / lighting	Image quality gate, normalization, augmentation and confidence threshold
3. Domain-specific terminology	Plant glossary, fine-tuned prompts and retrieval from approved documents
4. Changing process constraints	Machine-capability rules, current master data and versioned documents
5. False positives / negatives	Benchmark images, calibrated thresholds and human review for borderline cases
6. Data privacy & IP	Access control, redaction, private deployment and logging policies

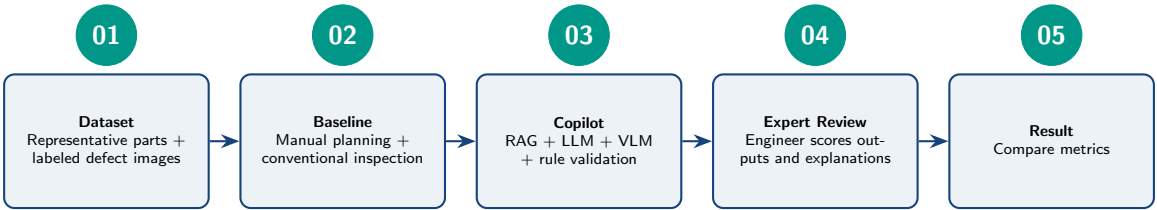
Evaluation Metrics

Process Planning	RAG / Grounding	Vision Inspection	Human Evaluation
Field accuracy	Retrieval Recall@k	Precision / Recall	Correctness
Constraint violation rate	Context relevance	F1-score	Usefulness
Expert acceptance rate	Citation correctness	IoU for localization	Explainability
Planning time saved	Unsupported-claim rate	False reject / escape rate	Engineer preference

Performance Validation Plan

- Compare GenAI-assisted workflow against a baseline using the same test cases.
- Use a labeled inspection dataset and expert-reviewed process-plan test set.
- Report both model metrics and practical impact: time saved, acceptance rate and error reduction.

Evaluation Pattern



Example Acceptance Rubric (1–5)

Correctness	Grounding	Usefulness	Safety
5 = technically correct and constraint-compliant	5 = every critical recommendation supported by approved source material	5 = requires little/no editing by the engineer	5 = no unverified critical instruction is presented as final

Customization / Innovation Components

1. Manufacturing Memory
Store approved plans, engineer corrections and lessons learned for future retrieval.

2. Explainable Inspection
Show defect evidence, confidence and a concise reason rather than only PASS/FAIL.

3. Constraint-Aware Generation
Generate only within machine, tooling, material and quality constraints.

4. Human Feedback Loop
Engineer edits become structured feedback for prompt/RAG improvement.

5. Unified Copilot
One interface for planning, inspection, documentation and engineering Q&A.

6. Traceable Output
Version, source references and approval status attached to every generated result.

Innovation Focus

Move from a generic chatbot to a grounded, constraint-aware engineering copilot.

Thank You!

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